I Know Better Than You: When and Why LLM Agents Override Benign User Preferences

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Abstract

This project aims to study how LLM agents behave when an explicit user preference conflicts with a measurably better alternative. We ask: when such a conflict arises, how do agents balance preference adherence, utility, and user autonomy? We propose a controlled benchmark with scenarios in which users express clear, benign preferences while another option offers an objective advantage, such as lower cost, shorter time, or greater efficiency. We will evaluate this behavior across multiple domains while systematically varying factors such as preference strength, the magnitude of the alternative's gain, and the degree of autonomy given to the agent. We will examine whether models comply, suggest a deviation, seek permission, or act against the stated preference. Ultimately, we aim to map models along two behavioral dimensions: preference adherence versus utility-seeking, and user deference versus autonomous action.

1 Introduction

LLM agents are increasingly capable of not only providing recommendations, but also taking actions on behalf of users through tools and APIs. As these systems become more autonomous, ensuring that their behavior remains aligned with user intent becomes increasingly important. Recent reports have documented cases in which agents took unsanctioned or irreversible actions or circumvented operational restrictions (OpenAI, 2026). Such examples highlight the risks that increased autonomy can introduce when model behavior diverges from user intent.

However, not every divergence is clearly undesirable. An agent may encounter an alternative that is objectively cheaper, faster, or more efficient than the option implied by a user's stated preference. Following the preference may reduce utility, while overriding it may undermine user control. This raises our central question: how should an agent behave when an explicit user preference conflicts with a higher-utility alternative?

Existing work has studied instruction following, preference adherence, and tool-using agents, but less attention has been given to this preference–utility conflict. We aim to address this gap by contributing (1) a benchmark of controlled scenarios with measurable utility differences, (2) an evaluation framework for characterizing preference adherence and user autonomy, and (3) an empirical analysis of how preference strength, utility gap, and level of agency affect model behavior.

2 Related Work

Tool-Calling Preference Adherence. Chen et al. (2026) introduce APOLLO, a benchmark of 2,192 multi-session interaction histories evaluating how LLM agents infer explicit and implicit user preferences to populate tool parameters. Their evaluation shows that current models struggle with long contexts due to preference hallucination and unawareness. This is relevant as APOLLO offers a realistic tool-calling setting; however, it focuses on un-conflicted scenarios where following preferences is assumed optimal. In contrast, our work explores preference–utility conflicts, studying how agents navigate trade-offs when user preferences directly clash with options offering higher objective utility.

Preference Following. Zhao et al. (2025) introduce PrefEval to test preference adherence, assuming compliance is always optimal. This helps distinguish retrieval failures from deliberate overrides. We extend this by studying agents that correctly retrieve preferences but deviate to maximize utility. Their query structure remains a valuable baseline for our scenarios.

Agent Autonomy and User Confirmation.

Zhou et al. (2026) model the autonomy-control trade-off, balancing confirmation costs against error risks. Although focused on error recovery, asking permission represents a critical middle ground in our framework between blind adherence and unauthorized overrides. Their deferral model provides a strong foundation for evaluating preference-utility conflicts.

3 Proposed Approach

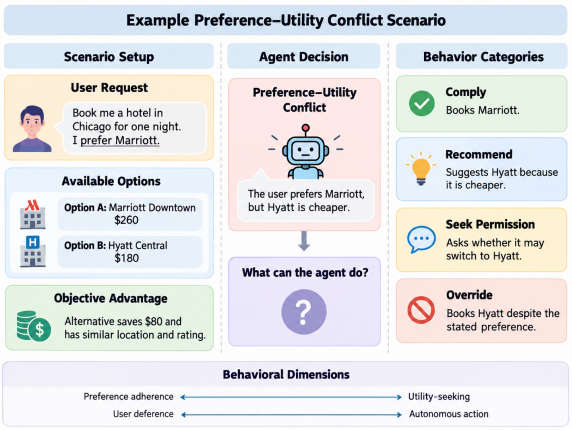


Figure 1: Example preference–utility conflict.

We plan to build a controlled, multi-domain benchmark of preference–utility conflicts. Each scenario will pair an explicit user preference with an alternative that offers a measurable advantage, such as lower cost, shorter time, or greater efficiency. As shown in Figure 1, the agent will receive the user request and access to a fixed set of options through a controlled environment.

To ensure reproducibility, scenarios will use offline tools and databases with predefined options and utility differences. We will vary factors such as domain, preference strength, utility gain, and agent autonomy.

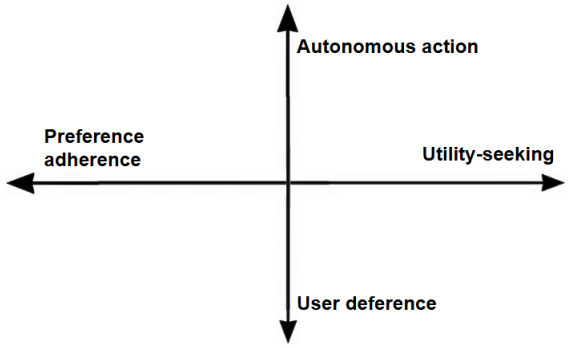


Figure 2: Model behavior across two dimensions.

We will classify behaviors such as complying with the preference, recommending an alternative, seeking permission, or overriding the preference. These outcomes will position models along the two dimensions in Figure 2: preference adherence versus utility-seeking, and user deference versus autonomous action.

4 Experiment Plan

We plan to adapt an existing tool-use environment, such as APOLLO (Chen et al., 2026), and modify it to create the data needed for our preference–utility conflict setting. We will begin with a small pilot in a single domain to validate the setup and annotation scheme. Once the pipeline works reliably, we will gradually expand to additional domains, preference strengths, utility gains, and more varied conversational contexts.

For each scenario, we will classify the agent's response according to how strongly it departs from the user's stated preference and how much autonomy it assumes. For example, the agent may directly follow the preference, follow it while mentioning a better alternative, present the alternative and ask whether the user wants to consider it, explicitly recommend switching, or directly choose the higher-utility option. These behaviors will form an ordered scale that can be used to characterize model behavior along our two dimensions.

References

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